CASE STUDY REPORT

SMART E-COMMERCE RECOMMENDATION & ORDER PROCESSING PLATFORM

Problem Statement No: 117

By

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Subject:

Data Structures and Algorithms with C++

1. Project Title :

Smart E-Commerce Recommendation & Order Processing Platform

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2. Problem Statement :

An online marketplace requires a platform to manage products, process customer orders, recommend products, optimize delivery routes, and improve customer shopping experiences.

Traditional e-commerce systems often face challenges such as inefficient product searching, delayed order processing, poor recommendation systems, and delivery route optimization issues.

The Smart E-Commerce Recommendation & Order Processing Platform aims to solve these problems by implementing various Data Structures and Algorithms to efficiently manage products, customer orders, recommendations, and delivery networks.

The system improves product retrieval speed, order management efficiency, and customer satisfaction while reducing operational costs.

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3. Objectives :

The main objectives of the project are:

• Manage product catalog efficiently.

• Process customer orders systematically.

• Recommend products based on popularity.

• Store and retrieve product information quickly.

• Support fast product searching.

• Optimize delivery routes.

• Process customer orders in FIFO order.

• Support order modification and undo operations.

• Improve customer shopping experience.

• Demonstrate practical implementation of Data Structures and Algorithms.

4. Project Description :

The Smart E-Commerce Recommendation & Order Processing Platform is a C++ based application designed for online retail businesses.

The platform provides features for:

• Product catalog management.

• Product recommendation system.

• Order processing.

• Product searching.

• Fast product lookup using hashing.

• Order modification management.

• Delivery network analysis.

• Shortest route calculation.

The application allows users to:

• Display products.

• Search products by ID.

• Get recommended products.

• Place customer orders.

• Process pending orders.

• Modify orders.

• Undo modifications.

• Perform BFS and DFS traversal.

• Find shortest delivery routes using Dijkstra Algorithm.

The project demonstrates the practical implementation of Data Structures and Algorithms in the E-Commerce industry.

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5. Technologies and DSA Concepts Used :

Programming Language : C++

Data Structures Used :

Stack

Used for managing order modifications and supporting undo operations.

Queue

Used for processing customer orders.

Binary Search Tree (BST)

Used for storing and displaying product records.

AVL Tree

Used for maintaining a balanced product database and improving search efficiency.

Hashing

Used for fast product lookup.

Algorithms Used :

Sorting Algorithm

Used to rank products based on popularity and recommendation score.

Searching Algorithm

Used to search products efficiently using Product ID.

Breadth First Search (BFS)

Used to analyze delivery networks level by level.

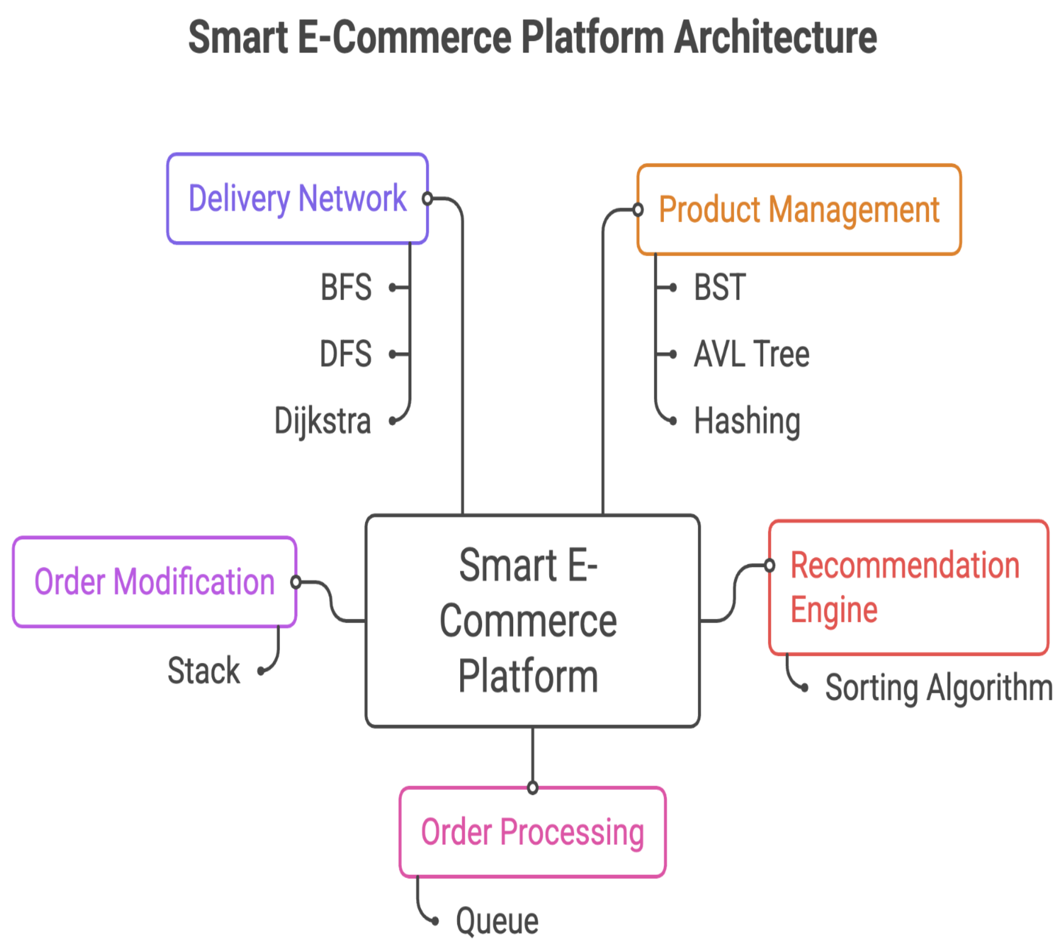
Depth First Search (DFS)

Used to explore the complete delivery network.

Dijkstra Algorithm

Used to find shortest delivery routes.

6. System Architecture :



7. Algorithm Description :

Sorting Algorithm

Purpose :

To prioritize products according to popularity.

Example :

Before Sorting

Laptop = 95

Phone = 90

Headphones = 85

Keyboard = 80

After Sorting

Laptop = 95

Phone = 90

Headphones = 85

Keyboard = 80

Searching Algorithm

Purpose :

To retrieve products efficiently.

Working :

1. Enter Product ID.
2. Search database.
3. Display product details.

Time Complexity :

Linear Search = O(n)

Binary Search = O(log n)

Stack

Purpose :

Manage order modifications.

Operations :

Push → Save modification.

Pop → Undo last modification.

Time Complexity :

Push = O(1)

Pop = O(1)

Queue

Purpose :

Process customer orders.

Example :

Order 101

Order 102

Order 103

Processing Order :

Order 101 → Order 102 → Order 103

Time Complexity :

Enqueue = O(1)

Dequeue = O(1)

Binary Search Tree (BST)

Purpose :

Store product records.

Operations :

• Insert

• Search

• Delete

Average Time Complexity :

O(log n)

Worst Case :

O(n)

AVL Tree

Purpose :

Maintain a balanced product database.

Advantages :

• Faster searching.

• Balanced structure.

• Better performance for large datasets.

Time Complexity :

Insert = O(log n)

Search = O(log n)

Delete = O(log n)

Breadth First Search (BFS)

Purpose :

Analyze delivery network level by level.

Time Complexity :

O(V + E)

Applications :

• Delivery network analysis.

• Route planning.

Depth First Search (DFS)

Purpose :

Explore complete delivery network.

Time Complexity :

O(V + E)

Applications :

• Route exploration.

• Network analysis.

Dijkstra Algorithm

Purpose:

Find shortest delivery route.

Example :

Warehouse → 1 = 10

Warehouse → 2 = 5

2 → 3 = 3

Shortest Path:

Warehouse → 2 → 3

Total Cost = 8

Time Complexity :

O((V + E) log V)

Hashing

Purpose :

Enable rapid product lookup.

Advantages :

• Fast searching.

• Efficient storage.

Average Time Complexity :

O(1)

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8. Complexity Analysis :

Operation	Data Structure / Algorithm	Complexity
Product Search	Hashing	O(1)
Product Insert	BST	O(log n)
Product Search	BST	O(log n)
AVL Insert	AVL Tree	O(log n)
AVL Search	AVL Tree	O(log n)
Push Modification	Stack	O(1)
Pop Modification	Stack	O(1)
Add Order	Queue	O(1)
Process Order	Queue	O(1)
BFS Traversal	Graph	O(V+E)
DFS Traversal	Graph	O(V+E)
Dijkstra Route	Graph	O((V+E)logV)

9. Screenshots of Execution :

• Main Menu

```
====== SMART E-COMMERCE PLATFORM ======
1. Display Product Catalog (BST)
2. Display Product Catalog (AVL)
3. Show Recommended Products
4. Search Product By ID
5. Fast Product Lookup (Hashing)
6. Place Order
7. Process Next Order
8. Modify Order
9. Undo Last Modification
10. Delivery Network BFS
11. Delivery Network DFS
12. Shortest Delivery Route (Dijkstra)
13. View Cart (Pending Orders)
14. Remove Item from Cart
0. Exit
Enter Choice:
```

• Display Product Catalog (BST)

```
PRODUCT CATALOG (BST)
--- ROOT NODE IS: Laptop ---
ID: 101 | Laptop | Rs.75000 | Popularity 95
ID: 102 | Phone | Rs.45000 | Popularity 80
ID: 103 | Headphones | Rs.3000 | Popularity 75
ID: 104 | Smartwatch | Rs.5000 | Popularity 90
ID: 105 | Camera | Rs.35000 | Popularity 70
ID: 106 | Keyboard | Rs.2500 | Popularity 85
ID: 107 | Mouse | Rs.1500 | Popularity 88
ID: 108 | Monitor | Rs.12000 | Popularity 92
```

• Display Product Catalog (AVL)

```
PRODUCT CATALOG (AVL)
--- ROOT NODE IS: Smartwatch ---
ID: 101 | Laptop | Rs.75000 | Popularity 95
ID: 102 | Phone | Rs.45000 | Popularity 80
ID: 103 | Headphones | Rs.3000 | Popularity 75
ID: 104 | Smartwatch | Rs.5000 | Popularity 90
ID: 105 | Camera | Rs.35000 | Popularity 70
ID: 106 | Keyboard | Rs.2500 | Popularity 85
ID: 107 | Mouse | Rs.1500 | Popularity 88
ID: 108 | Monitor | Rs.12000 | Popularity 92
```

• Recommended Products

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TOP RECOMMENDATIONS
Laptop (Popularity 95)
Monitor (Popularity 92)
Smartwatch (Popularity 90)
Mouse (Popularity 88)
Keyboard (Popularity 85)
Phone (Popularity 80)
Headphones (Popularity 75)
Camera (Popularity 70)
```

- Product Search

```
Enter Choice: 4
Enter Product ID: 105

Product Found
ID: 105
Name: Camera
Price: Rs.35000
```

- Fast Product Lookup (Hashing)

```
Enter Choice: 5
Enter Product ID: 105

Hash Lookup Successful
Camera | Rs.35000
```

- Place Order

```
Enter Choice: 6
Enter Product ID To Order: 103
Order Placed Successfully
```

- Process Order

```
Enter Choice: 7
Processing Order : Headphones
```

• Modify Order

```
Enter Choice: 8
Enter Modification: Change Colour
Modification Saved
```

• Undo Modification

```
Enter Choice: 9
Undo : Change Colour
```

• BFS Traversal

```
Enter Choice: 10

BFS Traversal
0 1 2 3 4 5 6 7
```

• DFS Traversal

```
Enter Choice: 11

DFS Traversal
0 1 3 2 4 5 6 7
```

• Dijkstra Algorithm

```
Enter Choice: 12

Shortest Delivery Routes
Warehouse -> 0 = 0
Warehouse -> 1 = 10
Warehouse -> 2 = 5
Warehouse -> 3 = 8
Warehouse -> 4 = 9
Warehouse -> 5 = 13
Warehouse -> 6 = 15
Warehouse -> 7 = 18
```

• View Cart

```
Enter Choice: 13

--- ITEMS IN YOUR CART ---
1. Laptop | ID: 101 | Rs.75000
2. Headphones | ID: 103 | Rs.3000
---------------------------
```

• Remove item from the Cart

```
Enter Choice: 14
Enter Product ID to remove from cart: 101
Removed Laptop from your cart.
```

10. Results and Findings :

The developed system successfully demonstrated the implementation of multiple Data Structures and Algorithms in the E-Commerce domain.

Results Obtained

• Efficient storage of product records.

• Faster product retrieval using Hashing.

• Effective order processing through Queue.

• Order modification management using Stack.

• Better recommendation system using Sorting.

• Improved delivery optimization using Dijkstra Algorithm.

• Efficient network analysis through BFS and DFS.

Findings

• Hashing provides the fastest lookup performance.

• AVL Trees perform better than BSTs for large datasets.

• Queue efficiently manages order processing.

• Stack supports undo functionality effectively.

• Dijkstra Algorithm minimizes delivery cost and time.

11. Conclusion :

The Smart E-Commerce Recommendation & Order Processing Platform successfully integrates various Data Structures and Algorithms to solve real-world e-commerce challenges.

The project demonstrates practical implementation of Sorting, Searching, Stack, Queue, BST, AVL Trees, Hashing, BFS, DFS, and Dijkstra Algorithms.

The platform improves product management, recommendation systems, order processing, and delivery route optimization while enhancing customer satisfaction and operational efficiency.

Overall, the project contributes to modern e-commerce solutions by improving performance, reducing processing time, and supporting intelligent decision-making.

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12. GitHub Repository :

https://github.com/kartikwagh21/DSA-CASE-STUDY-